

VIBHAV MANIKPURI

Jabalpur, Madhya Pradesh

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Education

PDPM IIITDM Jabalpur B.Tech in Mechanical Engineering	2023 – 2027
Madhya Pradesh Board of Secondary Education Higher Secondary Education	2021 – 2022

Technical Skills (Relevant to Motor Control R&D)

Control Systems (Classical Control)	MATLAB (Modeling, Analysis)
Electromechanical Systems	Sensors & Actuators
ANSYS (Structural & Thermal Analysis)	Mechatronics
SolidWorks (Mechanical Design)	Embedded Systems (Basics)
C / Python (Academic Use)	Data Analysis & Report Writing

Projects

SAEINDIA SUPRA 2025 — Formula Student Racing Car <i>Vehicle Dynamics Member & Driver</i>	2024 – 2025
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- Designed suspension and steering geometry using analytical vehicle dynamics and Ackermann steering principles.
- Modeled mechanical systems in SolidWorks and validated designs using structural stress analysis in ANSYS.
- Designed brake rotor and performed thermal and stress analysis to study heat dissipation and structural safety.
- Supported system integration and validation through manufacturing coordination and on-track testing.
- Prepared technical design and cost reports, emphasizing logical documentation and engineering justification.

AI-Assisted Smart Irrigation System (Design & Mechatronics Project)	2024 – Present
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- Designed mechanical architecture of an automated irrigation system integrating sensors and actuators.
- Built simulation models demonstrating sensor-based control logic using Tinkercad.
- Gained exposure to closed-loop control concepts through data-driven actuation behavior.

Exoskeleton Arm — Electromechanical Design Project	2024
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- Designed articulated mechanical structure of an exoskeleton arm in SolidWorks.
- Integrated servo motors with microcontroller-based control for motion actuation.
- Studied actuator response and mechanical load behavior during controlled movement.

Additional Experience

MP Government Hackathon — HACK & MAKE (Finalist)	2025
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- Developed engineering logic for cost modeling, system scalability, and real-world deployment.
- Worked in a structured, report-driven, problem-solving environment similar to industrial R&D.

Positions of Responsibility

- Vehicle Dynamics Cost Lead, SAEINDIA SUPRA 2025.
- Active Driver and Technical Member, Formula Student Team.